# Higher National Diploma in Engineering (Civil/QS)

## First Year, Second Semester Examination - 2019

## CE 1205/QS12044 - Civil Engineering Construction I

**Instructions for Candidate:**

Answer Five Questions Only

**No. of Questions : 06**
**No. of Pages : 03**
Time : 03 (hours)

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(01.)

I. Briefly explain the purpose of site investigation. (07 Marks)

II. What are the stages of site investigation? (03 Marks)

III. Define the following boring methods.
   a. Trial pits
   b. Auger boring
   c. Wash boring (06 Marks)

IV. Write down four no. of general considering factors in site construction initiation. (04 Marks)

(02.)

I. Briefly describe the following classification methods of earthwork.
   a. Confined excavation
   b. Deep excavation
   c. Embankment construction
   d. Bulk / open excavation (08 Marks)

II. State three types of rocks found in rock excavation. (03 Marks)

III. Briefly explain the difference between surface water and ground water. (04 Marks)

IV. Write down the methods of controlling ground water as a temporary and permanent solution. (05 Marks)

(03.)

I. State five functions of a foundation. (05 Marks)

II. What is the difference between Shallow and Deep foundations. (04 Marks)

III. Briefly explain two types of shallow foundations. (04 Marks)

IV. Define the term "safe bearing capacity of a soil". (03 Marks)

V. Consider a column with vertical load of 300 KN and safe bearing capacity of 150 KN/ $m^2$. Find out the safe area of the foundation. (04 Marks)

(04.)

I. State four functional requirements of walls. (04 Marks)

II. Brick is the most common material for constructing an external and internal wall in Sri Lanka. What are the advantages and disadvantages of bricks in wall construction. (04 Marks)

III. Briefly explain three types of bonding patterns of bricks with sketches. (06 Marks)

IV. As a Trainee Assistant Engineer, what are the considerable factors you should maintain in a wall construction. (03 Marks)

V. State three effects of dampness (03 Marks)

(05.)

I. List out five building services utilized in a hospital building. (05 Marks)

II. Name four factors which affects in selection of roof covering. (04 Marks)

III. State and compare the usage of three types of roof covering materials. (06 Marks)

IV. Briefly describe the construction process of brick paved and cement rendered floors. (05 Marks)

Temporary Elevated structure in ~ that they workers, tools, materials work in elev. are. unable to rea

(06.)

I. What is scaffolding? Explain its functions.

(03 Marks)

II. State the function of a door. What are the main three elements of a door.

(04 Marks)

III. List out four types of arches with sketches (based on the shape). coarse gothic

(04 Marks)

IV. Briefly explain the role of three parties involve in a construction industry.

(03 Marks)

V. Explain following activities with suitable equipment.

a. Excavation → scapper.

b. Earth compaction. → roller, plate.

(03x2 Marks)